

RAGINI SINGHAL

Curriculum Vitae

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Employment

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| Oct 2024-2027 | Postdoctoral researcher, University of Münster
Mentor - Hans-Joachim Hein
(funded by the Cluster of Excellence for the University of Münster). |
| Sep 2023-2024 | Research associate, Université Libre de Bruxelles
Mentor - Joel Fine |
| March-July 2023 | Visiting researcher, Humboldt-Universität zu Berlin
Mentor - Thomas Walpuski |
| Nov 2021-2022 | Research Associate, King's College London
Mentor - Simon Salamon
(funded by the Simons Collaboration on Special Holonomy). |

Education

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| 2017-2021 | Ph.D. in Mathematics, University of Waterloo , Waterloo, Canada.
Advisors - Benoit Charbonneau
Spiro Karigiannis |
| 2015-2016 | M.Sc. in Pure Mathematics, Imperial College London, UK .
M.Sc. Thesis - Stable minimal cones in Euclidean space.
M.Sc. Thesis Advisor - André Arroja Neves |
| 2011-2015 | BS in Mathematics, Indian Institute of Technology, Kanpur , India.
Undergraduate Thesis - Application of knot theory to detect chirality of molecules.
Undergraduate Thesis Advisor - Aparna Dar |

Research Interests

Riemannian geometry, Geometric analysis, Gauge theory, Deformation theory, Metrics with special holonomy, Lie groups, Spin geometry.

Preprints

- (2) S. Dwivedi, R. Singhal, Solutions and singularities of the Ricci-harmonic flow and Ricci-like flows of G_2 -structures [arXiv](#)
- (1) D. Festi, D. Platt, R. Singhal, Y. Tanaka, Examples of real stable bundles on K3 surfaces [arXiv](#)

Publications

- (4) S. Salamon, R. Singhal, Revisiting 3-Sasakian and G_2 -structures, **Ornea, L. (eds) Real and Complex Geometry Springer, Cham.** (2025) [arXiv](#) [Journal](#)
- (3) R. Singhal, *Nearly half-flat structures on $S^3 \times S^3$* , **Differential Geometry and its Applications** **97** [arXiv:2310.11233](#), [Journal](#).
- (2) R. Singhal, *Deformations of G_2 instantons on nearly G_2 manifolds*, **Annals of Global Analysis and Geometry** **62**, pages 329–366 (2022) [arXiv:2101.02151](#), [Journal](#).
- (1) S. Dwivedi and R. Singhal, *Deformation theory of nearly G_2 manifolds*, **Communications in Analysis and Geometry** Volume 31 Number 3 2023, [arXiv:2007.02497](#), [Journal](#).

In preparation articles

- (6) H.J. Hein, D. Ming, R. Singhal, *Seiberg–Witten theory on toric 4-manifolds*.
- (5) S. Salamon, R. Singhal, *$SO(4)$ -invariant nearly parallel G_2 -structures on Berger Space*.
- (4) B. Charbonneau, D. Harland, R. Singhal, *Higher order degormations of instantons on 6-dim Nearly Kähler and Nearly G_2 -manifolds*.
- (3) J. Fine, P. Ghosh, R. Singhal, *Seiberg–Witten equations on $Spin(7)$ -manifolds*
- (2) I. Alonso, S. Karigiannis and R. Singhal *Deformation of closed $SL(n, \mathbb{C})$ -structures with small torsion*
- (1) S. Dwivedi and R. Singhal, *Pluripotential theory and cohomology on $Spin(7)$ -manifolds*.

Invited Short-Term Visits

March-June 2024	CRM-Simons Research Visitor, CRM, Montreal, Canada
May-June 2014	Summer Research Visitor, Simon Fraser University, Canada
May-July 2013	Student Research Fellowship Program 2013, Indian Statistical Institute, New Delhi, India

Honours and Awards

- 2023 CRM-Simons Scholar at Centre de Recherches Mathématiques, March-June 2024.
- 2021 Doctoral Thesis Completion Award, University of Waterloo, Canada
- 2019 Graduate Studies Research Travel, University of Waterloo, Canada
- 2017-present Graduate Research Scholarship, University of Waterloo, Waterloo, Canada
- 2017-present International Doctoral Student Award, University of Waterloo, Canada
- 2017-present Provost Doctoral Entrance Award for Women, University of Waterloo, Canada
- 2015-2016 Imperial India Foundation Scholarship, Imperial College London, UK
- 2013 Summer Research Fellowship, Indian Academy of Sciences/ Indian National Science Academy/National Academy of Sciences, India
- 2011-2015 Kishor Vaigyanik Protsahan Yojana (KVPY) scholarship, Indian Academy of Sciences, Government of India

Teaching

Instructor & Coordinator

Spring 2020	MATH 117 - Calculus I for Engineers
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Teaching Assistant

Winter 2024	KG-2024 Kähler Geometry
Winter 2021	PMATH 450/650 Lebesgue integration and Fourier analysis
	PMATH 365/465 Differential geometry
Fall 2020	PMATH 331 Applied Real Analysis
	MATH 127 Calculus 1 for the Sciences
Winter 2020	PMATH 450/650 Lebesgue integration and Fourier analysis
	MATH 235 Linear Algebra (tutorial center)
Fall 2019	PMATH 365/465 Differential geometry
	MMT 647 Foundations of Calculus I
Spring 2019	PMATH 321 Non Euclidean geometry
	PMATH 351 Real Analysis
Winter 2019	PMATH 333 Introduction to real analysis
	MATH 118 Calculus 2 for Engineering
Fall 2018	MATH 235 Linear Algebra
	PMATH 331 Applied real analysis
Spring 2018	MMT 648 Foundations of Calculus II
Winter 2018	PMATH 365/465 Differential geometry
	MATH 235 Linear Algebra
Fall 2017	PMATH 365/465 Differential geometry
	MATH 147 Calculus I (Advanced level)

Services

- Organiser, Geometry, HOlonomy and Structures with Torsion (GHOST) 2026, 23 – 27 March 2026, Münster, Germany.
- Referee, Journal of Differential Geometry (International press).
- Referee, Quarterly Journal of Mathematics (Oxford University Press).
- Organizer, Workshop: Special Riemannian metrics in dimensions 6,7,8, Centre de Recherches Mathématiques, Montreal, 22-26 April 2024.
- Organizer, British Isles Graduate Workshop, Inverness, 3-7 July 2023.
- Organizer, Junior Special Geometers Meeting, King's College London, 6-8 January 2022.
- Volunteer instructor, [Shri Vidhya Jyoti Foundation](#), Jaipur, India.
- Volunteer, Mathematica Centrum Contest tutor, [K-W Bilingual School](#), Waterloo.
- Judge, Science Fair, [K-W Bilingual School](#), Waterloo.
- Co-organizer, Pure Mathematics Graduate Student Colloquium, University of Waterloo.
- Graduate Volunteer, The Great Polytope Barn-Raising Project, University of Waterloo.
- Grader, CEMC Math contest, University of Waterloo.

Invited Talks

- "Solutions and singularities of the Ricci-harmonic flow", Differential Geometry Seminar, 3 June 2026, University of Waterloo.
- "Cohomogeneity-one nearly G_2 -structures", Forschungsseminar über Differentialgeometrie, 6 January 2025, University of Hamburg, Germany.
- "Nearly half-flat structures on $S^3 \times S^3$ ", Mathematisches Seminar, 17 December 2024, University of Kiel, Germany.
- "Deformation theory of nearly G_2 -instantons" Geometric Models of Matter 2024, 21 August 2024, Leeds University, UK.
- "Cohomogeneity-one nearly G_2 -structures" Special geometries in Dimension 6,7,8, 26th April 2024, CRM Montreal
- "Deformation theory of G_2 -structures and instantons", Young Scholar day, 20 December 2023, Belgian Math Society.
- "Nearly half-flat structures on $S^3 \times S^3$ ", Oberseminar, November 27, 2023, Universität Münster.
- "Deformation theory of nearly G_2 manifolds", ULB Differential Geometry Seminar, May 2, 2023, Université Libre de Bruxelles.
- "Deformations of G_2 instantons on nearly G_2 manifolds", Oxford-London Gauge assembly, November 11, 2022 (online).
- "Deformations of G_2 instantons on nearly G_2 manifolds", Leeds geometry seminar, September 21 2022, University of Leeds.
- "Deformations of G_2 instantons on nearly G_2 manifolds", UCL geometry seminar, February 16 2022, UCL.
- "Deformations of G_2 instantons on nearly G_2 manifolds", Simons collaboration workshop "Connections between String Theory and Special Holonomy", January 10-14 2022, Oxford.
- "Deformation theory of nearly G_2 manifolds", Differential Geometry Seminar 2020-2021, UC Santa Barbara.
- "Deformation of instantons on nearly G_2 manifolds", AMS Fall Eastern Virtual Sectional meeting, 2020.
- "Deformation of instantons on nearly G_2 manifolds", Ottawa Mathematics Conference 2020, Ottawa.
- "Deformation theory of nearly G_2 manifolds", KCL/UCL Junior Geometry Seminar, 2020.
- "Deformation of instantons on nearly G_2 manifolds" **IIT Kanpur, India**; 11/11/2019
- "Deformations of G_2 instantons on nearly G_2 manifolds", G_2 geometry and related topics, **CMO-BIRS, Oaxaca, Mexico**; 07/05/2019
- "Minimal and Willmore surfaces", Graduate Student Colloquium, **University of Waterloo**, Canada; 26/03/2019

References

- Hans-Joachim Hein, University of Münster, (hhein@uni-muenster.de)
- Joel Fine, Université Libre de Bruxelles, (joel.fine@ulb.be)
- Simon Salamon, King's College London, (simon.salamon@kcl.ac.uk)
- Derek Harland, University of Leeds, (d.g.harland@leeds.ac.uk)
- Benoit Charbonneau, University of Waterloo (bcharbon@uwaterloo.ca)
- Spiro Karigiannis, University of Waterloo (karigiannis@uwaterloo.ca)